THE *VOYAGER* SPACECRAFT are bound for the stars. They are on escape trajectories from the Solar System, barreling along at almost a million miles a day. The gravitational fields of Jupiter, Saturn, Uranus, and Neptune have flung them at such high speeds that they have broken the bonds that once tied them to the Sun.

Have they left the Solar System yet? The answer depends very much on how you define the boundary of the Sun's realm. If it's the orbit of the outermost good-sized planet, then the *Voyager* spacecraft are already long gone; there are probably no undiscovered Neptunes. If you mean the outermost planet, it may be that there are other—perhaps Triton-like—planets far beyond Neptune and Pluto; if so, *Voyager 1* and *Voyager 2* are still within the Solar System. If you define the outer limits of the Solar System as the heliopause—where the interplanetary particles and magnetic fields are replaced by their interstellar counterparts—then neither *Voyager* has yet left the Solar System, although they may do so in the next few decades.* But if your definition of the edge of the Solar System is the distance at which our star can no longer hold worlds in orbit about it, then the *Voyagers* will not leave the Solar System for hundreds of centuries.

Weakly grasped by the Sun's gravity, in every direction in the sky, is that immense horde of a trillion comets or more, the Oort Cloud. The two spacecraft will finish their passage through the Oort Cloud in another 20,000 years or so. Then, at last, completing their long good-bye to the Solar System, broken free of the gravitational shackles that once bound them to the Sun, the *Voyagers* will make for the open sea of interstellar space. Only then will Phase Two of their mission begin.

Their radio transmitters long dead, the spacecraft will wander for ages in the calm, cold interstellar blackness—where there is almost nothing to erode them. Once out of the Solar System, they will remain intact for a billion years or more, as they circumnavigate the center of the Milky Way galaxy.

We do not know whether there are other spacefaring civilizations in the Milky Way. If they do exist, we do not know how abundant